

# Nagizaaz Shaik

Cloud Architect · ML Engineer · AI Engineer

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## SUMMARY

5+ years of production experience spanning AWS infrastructure design, model deployment pipelines, and LLM applications — working across Cloud Architecture, ML Engineering, and AI Engineering. At Wright-Patt Credit Union, designed and operates the full ML platform — five SageMaker-hosted production models, XGBoost underwriting scoring, BERT classifiers, RAG compliance tooling, and automated retraining — with zero findings across two consecutive compliance audits on \$400M in member financial data. Engineering portfolio covers three distinct domains: clinical pathology classification at 92% AUC (ML), LLM adversarial safety auditing against OWASP LLM Top 10 (AI), and privacy-preserving federated credit risk modeling across distributed institutions (Cloud). AWS ML Specialty certified.

## TECHNICAL SKILLS

**ML & AI:** PyTorch · XGBoost · BERT · HuggingFace Transformers · scikit-learn · LangChain · LangGraph · MLflow · AWS SageMaker · SageMaker Model Monitor · SHAP · Grad-CAM

**LLM & Agentic AI:** RAG Pipelines · Multi-Agent Orchestration (LangGraph) · OWASP LLM Top 10 · Red-Teaming · Adversarial ML · Federated Learning (Flower) · Presidio · FHIR R4

**Cloud & Infra:** AWS (SageMaker · EC2 · S3 · RDS · Glue · Redshift · IAM · CloudWatch · CloudTrail) · GCP (BigQuery · Vertex AI) · AES-256 · RBAC · HIPAA · ECOA

**MLOps:** Apache Airflow · MLflow Registry · SageMaker Pipelines · Docker · Kubernetes · FastAPI · CI/CD · A/B Testing · Automated Retraining · Model Versioning

**Data Engineering:** Apache Kafka · Spark / PySpark · dbt · AWS Glue · Redshift · Snowflake · Oracle SQL / PL/SQL · Informatica PowerCenter

**Languages:** Python · SQL · PL/SQL · Bash

## PROFESSIONAL EXPERIENCE

**Cloud Architect** at **Wright-Patt Credit Union** • Fairborn, OH

Jul 2024 – Jun 2026

- Designed the complete AWS ML platform architecture from scratch — SageMaker inference endpoints, MLflow model registry with version-gated promotion and automated rollback, VPC-isolated compute per environment, and IAM policies scoped to least privilege per service. Deployment failures reduced by 85% within 60 days.
- Defined and enforced the data security architecture across all ML workloads — AES-256 encryption at rest and in transit, CloudTrail logging on all inference calls, and VPC-isolated SageMaker endpoints. Passed two consecutive NCUA compliance audits with zero findings on \$400M in member financial data.
- Deployed an XGBoost credit underwriting model on SageMaker processing 500+ automated loan decisions per day — engineered the complete pipeline from AWS Glue feature preparation to real-time inference endpoint, with S3-backed structured logging for regulatory audit trail. Cut manual analyst review by 30%.
- Built production model monitoring using SageMaker Model Monitor with custom statistical baselines — daily scheduled jobs tracking feature distribution shifts and prediction confidence degradation across five live models. Intercepted two critical drift events before accuracy dropped measurably in production decisions.
- Replaced manual weekly model refresh with an Airflow DAG that retrains on a rolling data window, evaluates against holdout performance, and promotes via SageMaker Pipelines only when AUC, precision, and calibration stay within defined thresholds — eliminating 8+ hours of manual work per week.
- Fine-tuned BERT on internal support ticket data and deployed it as a FastAPI microservice on EC2 — 14 intent categories, sub-200ms p99 latency under production load, routing 10,000+ weekly member requests with a 40% reduction in misrouting rate.
- Built a RAG-based document QA system using LangChain and Pinecone, indexing 500+ internal policy documents and NCUA regulatory guidance — compliance analysts reduced average research time from over two hours to under 20 minutes per query.
- Mentored two engineers transitioning into ML — weekly code reviews and architecture sessions covering model evaluation, deployment patterns, and MLOps fundamentals. Both now independently own and ship production ML components.

**Data Engineer** at **Cognizant** • Hyderabad, India

Jul 2020 – Jun 2023

- Led the Oracle 11g to Amazon Redshift migration for a financial services client — redesigned schema for columnar storage with optimal distribution and sort keys, rewrote 20+ stored procedures as modular Airflow DAGs. Average analytical query runtime improved by 60% across a 25GB+ warehouse serving five business units.
- Designed fifteen production ETL/ELT pipelines in Apache Airflow across Oracle, SQL Server, flat files, and REST APIs, with a multi-stage validation layer running row-count reconciliation and null-rate checks at each hop. Surfaced a silent upstream corruption issue that had been invalidating three months of downstream financial reports.
- Built a real-time Kafka feature pipeline for a retail demand forecasting model — POS events aggregated in a five-minute tumbling window and written to Redshift for inference. End-to-end latency under 60 seconds contributed to a 28% reduction in emergency restocking orders.

- Refactored 20+ SQL and PL/SQL stored procedures — replaced correlated subqueries with CTEs, switched full table scans to index-range reads, and partitioned three high-volume fact tables. Nightly financial reporting job dropped from 3+ hours to under 15 minutes.
- Promoted to technical lead within two years — code reviews for three engineers, weekly data modelling sessions, and primary client contact across two accounts with 100% on-time delivery.

## PROJECTS

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### CheXpert Pathology Classifier ML Engineering ↗ [github.com/shaikn6/chexpert-pathology-classifier](https://github.com/shaikn6/chexpert-pathology-classifier)

Multi-label chest X-ray classification across all 14 classes of the CheXpert benchmark, achieving 92% AUC. Architecture combines DenseNet121 and EfficientNet-B4 in an ensemble with MC Dropout for uncertainty quantification on uncertain labels — a clinically important feature that flags low-confidence predictions for radiologist review rather than silently passing them through. Grad-CAM and ScoreCAM saliency maps provide region-level explainability overlays on the source image. Full DICOM ingestion pipeline with PHI scrubbing built for real-world clinical data handling.

*PyTorch · DenseNet121 · EfficientNet-B4 · MC Dropout · Grad-CAM · ScoreCAM · DICOM · AWS SageMaker*

### LLM Safety Auditor AI Engineering ↗ [github.com/shaikn6/llm-safety-auditor](https://github.com/shaikn6/llm-safety-auditor)

Automated red-teaming and safety evaluation framework for production LLMs. Executes 250+ adversarial attack vectors across six mutation strategies — jailbreak injection, role-playing bypass, indirect prompt injection, encoding obfuscation, context manipulation, and multi-turn escalation. Scores results against the full OWASP LLM Top 10 taxonomy, produces a calibrated risk score per vulnerability class, and generates a structured PDF audit report suitable for compliance review. Built for regulated-industry AI deployments where safety evidence is a requirement, not an afterthought.

*Python · HuggingFace · OWASP LLM Top 10 · Red-Teaming · Adversarial ML · Hallucination Detection · FastAPI · ReportLab*

### Federated Credit Risk Cloud & Distributed Architecture ↗ [github.com/shaikn6/federated-credit-risk](https://github.com/shaikn6/federated-credit-risk)

Three-institution federated credit risk modeling system using Flower FedAvg — zero raw data exchange between institutions. Each node trains locally on its own loan portfolio; only encrypted model weight deltas are shared. Includes an L2-norm model poisoning detection gate that rejects malicious gradient updates before aggregation, gradient clipping for training stability, and ECOA / Fair Lending compliance constraints applied to the global model to prevent discriminatory outcomes. Deployed as a containerised multi-node system with FastAPI coordination layer.

*PyTorch · Flower (FedAvg) · Federated Learning · Privacy-Preserving ML · ECOA Compliance · Docker · FastAPI*

## CERTIFICATIONS

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**AWS Certified Machine Learning – Specialty** · Amazon Web Services

**AWS Certified Cloud Practitioner** · Amazon Web Services

**GCP Professional Data Engineer** · Google Cloud

**Snowflake SnowPro Advanced: Data Engineer** · Snowflake

**dbt Analytics Engineering Certification** · dbt Labs

## EDUCATION

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### Master of Science, Computer Science · GPA 3.93 / 4.0

University of Dayton — Dayton, OH

Aug 2023 – May 2025

*Coursework: Advanced Machine Learning · NLP · Human-AI Interaction · Big Data Systems · Cloud Computing · Distributed Systems · Statistical Inference*

### Bachelor of Technology, Computer Science Engineering · GPA 8.9 / 10

CMR Engineering College — Hyderabad, India

Aug 2016 – May 2020

*Coursework: Data Structures & Algorithms · DBMS · OOP (Java) · Computer Networks · Operating Systems · Software Engineering · Design & Analysis of Algorithms*